

Web-based management and analysis of nature conservation measures in Schleswig-Holstein – current developments

Friedhelm Hosenfeld ¹, and Kay Krüger²

Abstract: The database for nature conservation measures (MDB) has been in operation since 2010 within the Environment Department of the State of Schleswig-Holstein as an internal web application for managing measures planned for nature conservation. This article describes recent key developments that have made the measures database application more user-friendly, more interoperable and better aligned with current requirements.

This includes a new module on the management of invasive alien species (IAS), which not only supports the implementation of EU requirements but also, beyond that, optimises the management of invasive species in Schleswig-Holstein.

Furthermore, the migration to the innovative Disy Cadenza Workbooks platform is discussed; whilst this platform offers comprehensive analysis and presentation options utilising linked dashboard displays, it also required the complete redevelopment of all existing analyses within the nature conservation measures database.

Keywords: nature conservation measures, invasive alien species, Disy Cadenza Workbooks, dashboard views, nature conservation management, OIIC

1 Introduction

The *nature conservation measures database* (MDB) described in this article was originally developed in 2010 as an internal web application for the nature conservation departments of the Ministry of the Environment (MEKUN) and State Agency for the Environment (LfU) in Schleswig-Holstein [Ho12].

The core application has been in use ever since, but has been continuously refined over the years, adapted to the latest technical requirements and optimised from a technical perspective.

A major change took place as early as 2019, when the application was migrated from the Oracle database system to PostgreSQL and operations were transferred from the LfU to the central twin data centre (TDC) of the state IT service provider Dataport. The switch to

¹ DigSyLand - Institute for Digital System Analysis & Landscape Diagnosis, Zum Dorfteich 6, D-24975 Husby, hosenfeld@digsyland.de, <https://orcid.org/0000-0002-0538-7684>

² MEKUN - Ministry for Energy Transition, Climate Protection, Environment and Nature Schleswig-Holstein, Mercatorstraße 3, D-24106 Kiel, kay.krueger@mekun.landsh.de

the open-source database PostgreSQL was carried out based on Schleswig-Holstein's open-source strategy [MP24].

This article discusses further extensive modernisations and enhancements that have recently been carried out or are currently being implemented, covering both technical aspects and subject-specific requirements:

- In terms of content, the measures database has been expanded to include a module for the management of invasive species, which not only covers the introduction of a new category for specific measures against invasive species but also takes invasive species into account within other categories of measures.
- As a major technical innovation – one which also opens up new technical possibilities in terms of analysis capabilities – a migration was carried out to the Disy Cadenza *Workbooks* analysis and GIS platform, having previously used the predecessor solution, Cadenza *Classic*.

Section 2 below outlines the current situation and the requirements for the latest developments, beginning with the introduction of the nature conservation measures database as a web application. The same section then sets out the rationale for the required enhancements and explains the requirements for both key areas of development.

Section 3 then examines the design and implementation of measures to manage invasive species. The final focus in Section 4 is on describing the migration from Cadenza Web Classic to Cadenza Workbooks, including the implementation of specialist analysis requirements in modern dashboards and the integration of the management application for data maintenance into Cadenza Workbooks. Finally, Section 5 provides a summary and outlook.

2 Background, motivation and requirements

2.1 Nature Conservation Measures Database (MDB) in Schleswig-Holstein

As the first implementation phase of a cross-cutting data management concept developed in 2009 for all nature conservation data within the environmental department, the nature conservation measures database (MDB) was developed in 2010 on behalf of what is now the Ministry for Energy Transition, Climate Protection, Environment and Nature (MEKUN), as until then no suitable tool for managing measures had been available within the state authorities.

From a technical standpoint, the MDB was implemented as a web application using the PHP scripting language (<https://www.php.net/>) and integrated into the Disy Cadenza analysis platform [DI26]. The PHP management application handles the recording, maintenance, and management of nature conservation measures, while analyses and map

visualizations were implemented in Cadenza Web. Furthermore, the GIS-based capture of measure geometries was one of the first applications realized in Cadenza Web using the Cadenza application interface [Ho12].

The application is used by over 80 staff members from the nature conservation departments of the LfU and MEKUN to manage currently more than 16,000 measures. Various categories of measures are distinguished – examples include species conservation measures, protection and development measures, and biotope creation measures. Access rights are determined based on two factors: the origin of the measure and the responsible authority (LfU or MEKUN).

The data is stored in a relational database management system (RDBMS) based on PostgreSQL and PostGIS at the central twin data centre of Dataport, the state's IT service provider (see Figure 1). This applies equally to two other key applications in the nature conservation sector: the Register of Protected Areas (PAR) and the Peatland Database, both of which are closely linked to the MDB. The Protected Area Register records all nature conservation-relevant protected areas, such as nature conservation areas, bird sanctuaries, and other Natura 2000 sites [Ho21]. The Peatland Database, in turn, not only records basic data on peatlands and specific peatland-related measures but also integrates all measures managed in the MDB that pertain to peatlands, thereby avoiding duplicate entries [Ho23] (see Figure 1). Both the PAR and the Peatland Database share the same technical architecture as the MDB, utilizing PHP data maintenance applications integrated into Cadenza.

2.2 Need for the management of invasive species

The importance of combating invasive species was recognized years ago and is increasingly gaining public attention. As early as 2008, the EU estimated invasive alien species were having a massive negative impact on the economy and the environment [EU09] which led in 2014 to EU Regulation 1143/2014 on the prevention and management of the introduction and spread of invasive alien species [EU14].

This regulation has two primary objectives:

- *prevention*: averting the introduction and spread of species that threaten biodiversity
- *control*: detecting such species that have already spread as early as possible, and subsequently eradicating them or, at the very least, monitoring and regulating them

In addition to collecting data on invasive species and implementing measures, the EU regulation requires Member States to report on species occurrences and the measures taken. The nature conservation departments of MEKUN and LfU worked on a technical concept deciding that data collection should take place within the existing nature conservation measures database in order to fulfill the needs of EU Regulation No. 1143/2014 and further local requirements concerning invasive alien species in Schleswig-Holstein.

2.3 Fundamentals of the Disy Cadenza Analysis and GIS Platform

Disy Cadenza is a comprehensive analysis platform offering a wide range of options for querying and visualizing non-spatial data, as well as map-based analyses. A particular strength lies in its functionality for the combined analysis and visualization of non-spatial and geospatial data; these features are tailored to the specific requirements of the environmental sector and are well-suited for the MDB (see also [DI26]). The universal analysis functions – which can be combined like building blocks and have proven highly effective in the field of environmental data – were a key reason for using Cadenza for the MDB.

Cadenza enables the seamless integration of dedicated management applications for data maintenance, while Cadenza itself focuses on providing read-only access for analysis purposes. Consequently, it serves as a modular system for designing analyses and visualizations (tables, charts, maps, reports, dashboards) based on standardized components, while a specialized management application tailored to the specific use case handles data maintenance – yet to the end-user, both components appear as a single, unified application (details on the current Cadenza Workbooks can be found in [JA26]).

2.4 Reasons and requirements for modernizing analyses and switching to Cadenza Workbooks

Cadenza Workbooks is designed to empower experts to customize dashboards and analysis views according to their needs, reducing their reliance on the centrally defined configurations typical of the earlier Cadenza Classic variant. Furthermore, the system's more flexible analysis and visualization options – particularly the ability to combine different views within dashboards – are driving the migration of existing applications to Cadenza Workbooks, like the MDB web application. While Cadenza Workbooks offers additional, useful analysis capabilities to optimize MDB data analysis, Cadenza Classic will receive support and further development for only a limited time; these were compelling reasons to switch to Cadenza Workbooks.

Analyses compiled in Cadenza for a specific business context are referred to as a "repository." The migration from Cadenza Classic to Cadenza Workbooks necessitated the complete reconstruction of all MDB related analyses and visualizations within the new environment due to the differing technical foundations. The new MDB workbooks repository was developed entirely by the responsible expert at MEKUN.

In contrast, the core MDB management application itself could continue to be used. Nevertheless, two significant adjustments were required to integrate the MDB management application with Cadenza Workbooks:

Since Cadenza Workbooks does not have its own user management system but instead relies on a central identity management solution like Keycloak via OpenID Connect (OIDC), a corresponding OIDC integration had to be established for the MDB management application (see Figure 1).

In addition, the management application utilized the geometry capture functionality previously provided by Cadenza Web Classic; this interface had to be migrated within the PHP application to the new Cadenza API provided by Cadenza Workbooks.

3 Design and implementation of the expansion to include invasive species

3.1 Technical aspects regarding Invasive Alien Species (IAS) data

As part of the requirements analysis, decisions were made regarding the necessary data fields, new reference lists, and the potential integration of the new invasive species module.

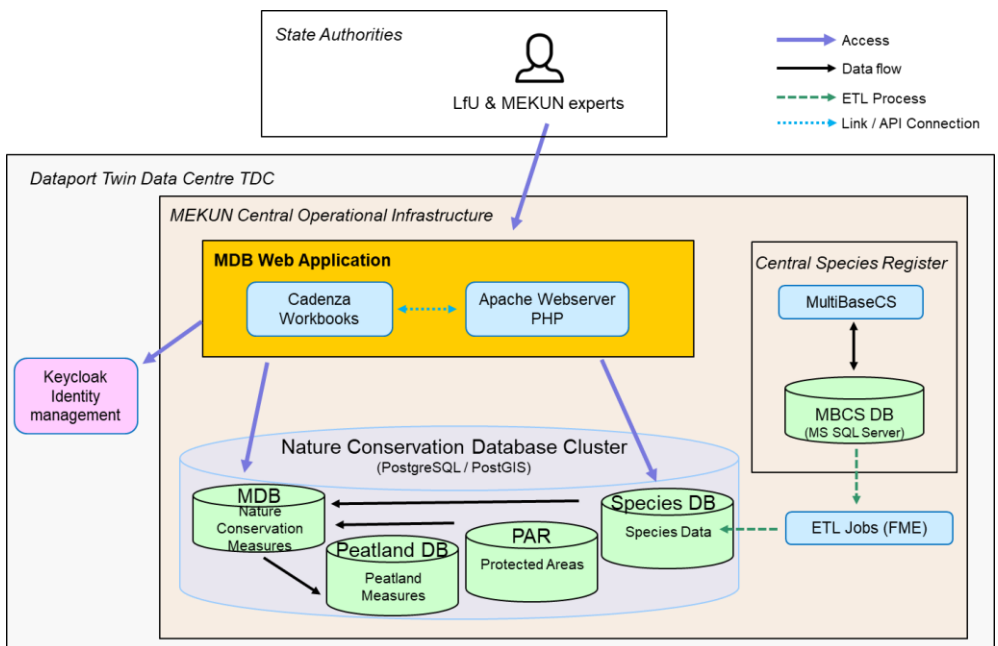


Figure 1: Main components used by MDB web application including key data flows, links, and access paths

A new measure category designated as "IAS measures" was introduced to capture all measures specifically planned to address invasive species that do not fall under other measure categories. Additionally, for other categories – such as conservation and development measures – a question was added to determine whether they also target invasive species. If so, the input fields for the respective measure category are supplemented with the additional fields required for invasive species management.

Monitoring impacts is essential for evaluating the success of invasive species control efforts and should be conducted at specific milestone dates. To accommodate this so-called *milestone monitoring*, another measure category designated as "monitoring measure" was introduced. This allows multiple monitoring measures to be linked to a single IAS measure, documenting the measure's success at the respective milestone dates.

3.2 Linking to specific occurrence locations of invasive species

A key requirement for measures addressing invasive species was establishing a data link between the actual occurrences of invasive species and the location data managed in the Central Species Register implemented using MultiBaseCS software (<https://www.multibasecs.de/>) [Ar24]. Consequently, each IAS measure in the MDB can be linked to one or more species occurrence records from MultiBaseCS. Since MultiBaseCS operates in a different technical environment, a direct link between the two applications could not be established. However, species data from the MultiBaseCS database is regularly transferred via ETL process (Extract – Transform – Load) to the nature conservation database, which is also used by the MDB (see Figure 1). This data also forms the basis for the species list within the MDB.

The MDB offers two methods for specifying the occurrence of an invasive species to which an IAS measure relates: either entering the data ID of the occurrence location from MultiBaseCS, or by selecting the occurrence location on the map.

For direct entry of the MultiBaseCS data ID, a filterable selection list of species occurrences is provided; this allows for direct ID entry while displaying the year and species name. This method is suitable for specialists who work in MultiBaseCS concurrently and can transfer the ID directly from that system.

When selecting via the map, the relevant species can be specified beforehand; this ensures that only occurrences of that specific species are displayed on the map and made available for selection and inclusion in the measure.

Additionally, every measure in the MDB includes one or more measure geometries, utilizing point, line, or area geometry types. A transfer function can be used to automatically adopt the point geometry of the species occurrence from MultiBaseCS as the measure geometry.

4 Migration from Cadenza Web Classic to Cadenza Workbooks

4.1 Setting up a new Workbooks repository with a dashboard and interactive analyses

The analyses compiled for the MDB – the MDB repository in Cadenza Classic – consisted of approximately 60 "filter forms" used to run queries; these typically presented search results as tables, but also as reports, charts, and maps.

The differing technical architecture of Cadenza Workbooks meant that all repository elements had to be rebuilt from scratch, as automatic migration was not possible. This reconstruction process provided an opportunity to align the analyses – which had evolved organically since their introduction in 2010 – with current requirements and to review them in light of the extensive new visualization options introduced with Cadenza Workbooks.

The new MDB repository developed by MEKUN in Cadenza Workbooks comprises 14 "workbooks" (see Figure 2), each containing multiple "worksheets."

The worksheets within a workbook are often linked by shared filter criteria, enabling different perspectives on a search (e.g., switching from an overview display to a detailed view). A worksheet can contain a dashboard comprising various individual views – such as tables, charts, indicators, and maps – as well as explanatory text with embedded current MDB data (see Figure 2). Key metrics – serving as so-called indicators – can be highlighted on a dashboard based on research findings, making the most important facts graspable at a glance. These newly available presentation options made it possible to tailor the analyses precisely to the functional requirements for using the MDB.

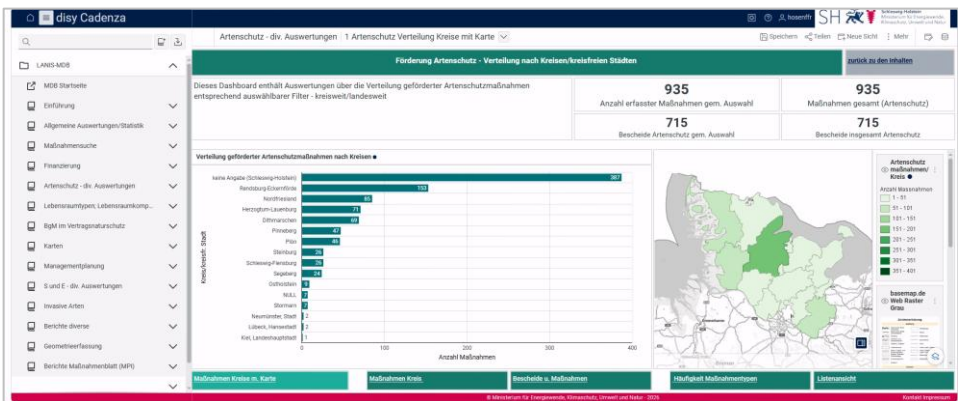


Figure 2: Cadenza „worksheet“ featuring a dashboard which combines various views displaying diagrams, tables, indicators, and maps (here: fictitious test data); the 14 „workbooks“ are listed on the left-hand side

4.2 Migration of the MDB Nature Conservation Management Application

The MDB management application, implemented in PHP, required adjustments for integration into Cadenza Workbooks regarding the following aspects:

- Connection to a central identity management system based on OpenID Connect, involving modifications to session management and access rights groups
- Utilization of the geometry capture functionality provided by Cadenza Workbooks via the new Cadenza API (`cadenza.js`)
- Addition of entry points to enable navigation from the Workbooks repository into the management application

In Schleswig-Holstein, the Keycloak software (<https://www.keycloak.org/>) is used for central identity management at the state data centre (see Figure 1). This Keycloak solution enables the connection of user credentials for all state agency employees via the statewide Active Directory (AD) while also allowing for the integration of external users through a unified interface.

The previous direct connection of the MDB to Cadenza's user management had to be replaced by an OpenID Connect (OIDC) interface to Keycloak. A recently developed suitable OIDC interface for another PHP-based management application in the geology sector [Ho25] – also created for the environmental department – could be reused and adapted to meet the specific requirements of the MDB application. It ensures that the user groups determining access rights within the MDB are appropriately mapped in both Keycloak and Cadenza and offers the convenience of Single Sign-On via Keycloak.

Another necessary adjustment involved reconfiguring the geometry capture function to use the Cadenza Workbooks API. The new JavaScript-based Cadenza API offers significantly more flexible integration and control options. However, it provides only client-side geometry capture. Consequently, the management application had to be extended with its own functionality for validating and saving geometries in the PostgreSQL (with PostGIS) database system. An advantage of this approach is that geometry capture can be embedded directly into the PHP-based management application.

Another benefit of switching to Cadenza Workbooks for the MDB is the availability of enhanced integration options between Cadenza dashboards and the management application; specifically, "drill-through" functions allow users to navigate directly from any Cadenza analysis view to the corresponding data entry forms in the PHP-based management application. This enables users to perform complex searches based on domain-specific criteria within Cadenza and then sequentially open and process all matching measures in the management application.

5 Summary and outlook

The nature conservation measures database (MDB) has been an important tool for the state authorities in Schleswig-Holstein since 2010, enabling them to plan, manage and monitor nature protection measures. The described recent enhancements and modernization efforts ensure continued efficient support for nature conservation tasks.

The addition of an invasive species module fulfills not only the recording and reporting obligations of EU Regulation No. 1143/2014 on invasive species [EU14] but also addresses additional requirements for invasive alien species (IAS) management in Schleswig-Holstein. Not only a dedicated category for IAS measures was added as well as a monitoring measure category in order to document results in combating alien species, but the IAS details could also be provided for measures of other categories.

As a further step in modernization, the MDB is being migrated from the previous analysis and GIS platform, *Cadenza Classic*, to the current web-based variant, *Cadenza Workbooks*. By rebuilding all analyses to be optimized for Cadenza Workbooks, the MDB's extensive data is being made accessible in a modern way and presented clearly. The new repository of MDB related analyses and visualizations unlocks newly available options and forms of presentation, such as dashboards and interconnected worksheets. The necessary OpenID Connect interface added to the PHP management application enables the user to utilize Single Sign-On. Greater flexibility regarding the integration of Cadenza and the management application is achieved by the Cadenza Workbooks API – for example to edit the geometries of measures.

Plans for the near future include optimising the integration of the PHP based management application into Cadenza Workbooks, as well as further expanding the analysis functions by utilising the new display formats available in Cadenza Workbooks.

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